João Pedro da Conceição Simas

Fullstack Engineer — Python · TypeScript · AI Systems

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SUMMARY

Fullstack engineer with 4+ years of hands-on experience and 1 year shipping production systems at scale. Currently sole engineer on a Python/FastAPI ERP — **2,190 commits across 30+ modules in 11 months** — including an LLM-powered WhatsApp agent, facial biometric authentication, and bank statement parsers for 5 Brazilian banks. Strong background in systems design, AI integration and CI/CD automation.

EXPERIENCE

Fullstack Engineer — AV09 Foreign Trade *Itajaí, BR · Apr 2025 – Present*

- Built a **production WhatsApp AI agent**: deterministic regex bypass before LLM call, normalization of invisible Unicode marks injected by WhatsApp Web, opaque LID resolution for Evolution API v2, 13 admin tools, Whisper transcription and web search.
- Designed statement parsers for **5 Brazilian banks** (BB, Bradesco, Itaú, Santander mTLS+OAuth2, Sicoob) with SHA-256 deduplication, multi-currency support and AI-assisted PDF parsing.
- Implemented **facial biometric auth** using MTCNN + FaceNet/InceptionResnetV1 with 4-criteria simultaneous matching: cosine similarity, Euclidean distance, separation margin and cohort scoring against top-5 similar faces.
- Architected **CI/CD pipeline from scratch**: infrastructure file-change guard, automated PR review, self-hosted runners — reduced deploy time from **35 minutes to 33 seconds**.
- Architected and developed an **integration middleware in .NET/C#** between the company's in-house ERP and **TOTVS Protheus**, structured in layers (Domain, Application, Infrastructure, API). Communication relies on an asynchronous queue with concurrency control, OAuth2 authentication, a retry policy with circuit breaker and failover, adaptive timeouts calibrated by each operation's historical p99, and confirmation callbacks to the originating system.
- Integrated Protheus with **payroll and onboarding, inter-branch transfers, shipping manifests, cost centers, product and customer registration, sales orders and receivables settlement**, as well as accounting reconciliation between journal entries and balances.
- Delivered over 30 backend modules in Python/FastAPI, including an audit system covering more than 210 routes, delivery routing with the Valhalla truck engine, multi-source exchange rate quotation and server metrics history.

Studies & Personal Projects *Remote · Apr 2024 – Mar 2025*

- Deepened AI/ML and software engineering skills — CS50's Introduction to Artificial Intelligence with Python and CS50's Introduction to Programming with Python (HarvardX) — and built open-source personal projects (Cashly, Prozes, among others), publishing a package on PyPI.

Fullstack Engineer (Freelance) *Remote · Jun 2022 – Mar 2024*

- Delivered web, automation and data science projects for clients across multiple industries under NDA.
- Managed systems serving **tens of thousands of active users**; significantly cut critical process times and operational costs through infrastructure optimisation.

SELECTED PROJECTS

Cashly · *Kotlin · Jetpack Compose · Supabase Realtime · TypeScript · React* · cashly.jaopd.dev

Android finance app that captures transactions automatically by reading bank push notifications in real time. Custom notification parser with intelligent categorisation and continuous learning, offline-first sync via WebSocket Realtime, biometric auth and recurring transactions with AlarmManager.

Prozes · *Python CLI · Published on PyPI* · prozes.jaopd.dev

Scaffolds production-ready Flask / FastAPI / Django projects in seconds, with test cases, virtualenv setup and CI/CD workflows baked in. Multilingual; open source.

RescueStick · *C++ · Python · Linux* · rescue-stick.jaopd.dev

Professional bootable USB system (Debian Linux) for IT technicians, with 7 tools in C++ and Python: Windows password reset, disk backup and recovery, hardware diagnostics, boot repair, file recovery and malware scanning. Modular architecture, MIT license and 100% test coverage.

TECHNICAL SKILLS

Languages	**Python**, **TypeScript**, JavaScript, Kotlin, C#, PHP, C/C++, Bash
Frontend	React, React Native, Tailwind CSS, Next.js
Backend	**FastAPI**, Node.js, Express, ASP.NET Core, Django, Flask, TOTVS Protheus integration
Databases	PostgreSQL, MySQL, MongoDB, Redis, SQL Server, Supabase, Firebase
DevOps	Docker, Linux, Git, GitHub Actions, GitLab CI, AWS, Azure, Cloudflare
AI / ML	OpenAI API, Anthropic API, MTCNN, FaceNet, TensorFlow, OpenCV, Whisper, Scikit-learn

EDUCATION & CERTIFICATIONS

Technologist in Systems Analysis & Development — UNINTER, Brazil *1st of 4 semesters · In progress*

- CS50's Introduction to Artificial Intelligence with Python — HarvardX (edX)
- CS50's Introduction to Programming with Python — HarvardX (edX)
- ASP.NET Core MVC — Entra21

LANGUAGES

Portuguese (native) · English (C1) · Spanish (B2) · Italian (B1)